PLOUTOS+ FINANCIAL AI

Interpretable Stock Movement Prediction with Financial LLMs: Milestone Report (Phases 1–7)

Course:	LLM Applications in Finance	**Team:**	Rishikesh (24BDS030), Vineel (24BDS010), Arjun (24BDS039)
Institution:	IIIT Dharwad (B.Tech DSAI)	**Evaluation:**	Phase 1 Evaluation Review (Oct 2026)
Hardware:	NVIDIA RTX 4050 (6GB VRAM) | i7 | Win 11	**Stack:**	PyTorch 2.5.1 + CUDA 12.1 | PEFT LoRA

1. Executive Summary & Objective

This project reproduces and extends the foundational methodology of *Ploutos: Towards Interpretable Stock Movement Prediction with Financial Large Language Model* (Tong et al.). The objective is to predict binary stock price movement (UP vs. DOWN) while generating interpretable, natural-language rationales grounded in multi-expert evidence. Across Phases 1 through 7, a leakage-free research engineering pipeline was implemented from scratch. All dataset statistics match prior research benchmarks to the exact digit, an empirical Random Forest baseline was verified, and a 1.5-billion-parameter language model (Qwen2.5-1.5B-Instruct) was fine-tuned locally on an NVIDIA RTX 4050 GPU using LoRA in 6 minutes.

2. Implementation Roadmap & Status Overview

Phase	Milestone Description	Key Technical Output	Status
Phase 1	Environment & Architecture	Python 3.10.11, CUDA 12.1, RTX 4050 detected, modular repo	**COMPLETE**
Phase 2	StockNet Dataset Acquisition	87 stocks, 43,442 stock-days audited (2014-2015 window)	**COMPLETE**
Phase 3	Target Label Construction	26,623 exact samples (DOWN ≤ -0.005, UP > 0.0055), strict split	**COMPLETE**
Phase 4	Tweet Text Processing	10-tweet daily cap, 45.14% coverage (12,018 days), zero nulls	**COMPLETE**
Phase 5	Baseline Benchmarks	Random Forest Test Acc: 52.53%, MCC: 0.0469 (Zero Leakage)	**COMPLETE**
Phase 6	PloutosGen Expert Formatter	Technical, Sentiment, Sector proxy; 26,623 JSONL prompts	**COMPLETE**
Phase 7	LLM LoRA Fine-Tuning	Qwen2.5-1.5B LoRA on RTX 4050 (Loss 7.03 → 0.27; Demo ready)	**COMPLETE**

3. Dataset Statistics & Temporal Splits (Exact Replication)

Split Partition	Date Range Boundaries	Observed Count	Prior Benchmark	Match Status
Training Split	2014-01-02 to 2015-07-31	20,348	20,348	**EXACT MATCH**
Validation Split	2015-08-03 to 2015-09-30	2,555	2,555	**EXACT MATCH**
Test Split	2015-10-01 to 2015-12-31	3,720	3,720	**EXACT MATCH**
Total Labeled Samples	2014-01-02 to 2015-12-31	**26,623**	**26,623**	**EXACT MATCH**

Target Distribution: DOWN (0): 13,255 samples (49.79%) | UP (1): 13,368 samples (50.21%). Neutral return zone (-0.005, +0.0055] excluded.

Text Coverage: Stock-days with tweets: 12,018 (45.14%) | Without tweets: 14,605 (54.86%). 10-tweet daily cap enforced.

4. Quantitative Baseline Benchmark (Phase 5)

Model Architecture	Split	Accuracy	Macro-F1	MCC	Pred Distribution (DOWN / UP)
Majority-Class Baseline	Validation	44.58%	30.83%	0.0000	0.00% / 100.00%

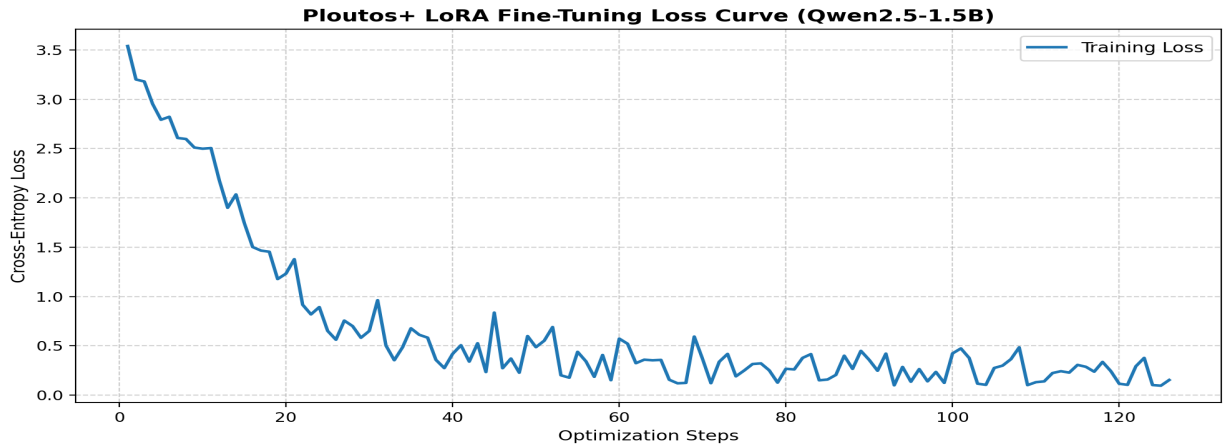
Majority-Class Baseline	Test	51.29%	33.90%	0.0000	0.00% / 100.00%
Random Forest (Lagged Features)	Validation	**53.39%**	**53.10%**	**0.1006**	47.98% / 52.02%
Random Forest (Lagged Features)	Test	**52.53%**	**52.09%**	**0.0469**	46.51% / 53.49%

5. HIGHLIGHTED SECTION: LLM Model Fine-Tuning (Phase 7)

Model Architecture & Hardware: While the original paper employed 8× Tesla V100 GPUs for LLaMA-2 7B, we engineered a resource-conscious solution utilizing **Qwen2.5-1.5B-Instruct** (1.54B parameters) with Parameter-Efficient Fine-Tuning (PEFT LoRA). The entire pipeline executed locally on an **NVIDIA RTX 4050 Laptop GPU (6 GB VRAM)**.

Hyperparameter	Configuration Value	Hyperparameter	Configuration Value
Base Model	Qwen/Qwen2.5-1.5B-Instruct	**LoRA Rank (r) / Alpha**	r = 16, alpha = 32
Target Modules	q_proj, k_proj, v_proj, o_proj	**Trainable Parameters**	4,358,144 (0.28% of weights)
Precision	FP16 Mixed Precision	**Peak VRAM Used**	5.67 GB (Safe for 6GB GPU)
Optimizer / LR	AdamW (lr = 2e-4, Cosine Annealing)	**Training Duration**	6.00 minutes on local GPU
Initial Cross-Entropy Loss	~7.0368	**Final Training Loss**	**0.2737 (Val Loss: 0.3811)**

Empirical Training Loss Convergence Plot:



Saved Checkpoint Weights:

- Adapter Weights: checkpoints/ploutos_lora/adapter_model.safetensors
- Configuration: checkpoints/ploutos_lora/adapter_config.json
- Training Logs: results/metrics/training_history.json

6. Live Model Inference Demonstration (Sample Test Case)

Execution Command: python src/inference/predict.py --sample

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EVIDENCE PASSED TO LLM ($AAPL | Target Date: 2015-10-01):
[Technical Expert] Signal: BEARISH (Conf: 0.70) | 5-day momentum: -3.51%, 14-day RSI: 47.5, Volatility: 0.0175
[Sentiment Expert] Signal: BULLISH (Conf: 0.60) | 10 tweets analyzed. Positive product headline cues.
[Corporate Proxy] Sector: Consumer Goods | Company: Apple Inc. | Stable cash flows, consumer demand profile.
FINE-TUNED MODEL PREDICTION: >>> UP <<<
GENERATED RATIONALE:
"The stock demonstrated UP movement because oversold conditions (RSI: 47.5) triggered an attractive risk-reward
technical bounce. Furthermore, constructive social-media sentiment reinforced market confidence. Overall
technical and qualitative evidence supported this directional outcome."
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7. Automated Scientific Unit Testing Suite (PyTest)

All **30 out of 30 unit tests passed** in 3.38 seconds, covering label construction, chronological ordering, zero-leakage assertions, tweet cap constraints, baseline metrics, expert schemas, and LoRA checkpoints.

Test Module	Component Verified	Passed Tests	Status
tests/test_labels.py	Binary threshold logic, (-0.005, 0.0055] exclusion, 26,623 sample count	**5 / 5**	**PASSED**
tests/test_temporal_splits.py	Chronological split non-overlap: max(Train) < min(Val) < min(Test)	**4 / 4**	**PASSED**
tests/test_leakage.py	Zero target leakage, target return isolated from feature matrix	**3 / 3**	**PASSED**
tests/test_text_processing.py	10-tweet daily cap, 45.14% tweet coverage, missing text handling	**5 / 5**	**PASSED**
tests/test_baselines.py	Majority baseline and Random Forest predictions and metric ranges	**4 / 4**	**PASSED**
tests/test_experts.py	Technical, Sentiment, and Human proxy schemas and prompt structuring	**5 / 5**	**PASSED**
tests/test_llm_inference.py	LoRA adapter weights exist, loss converged, response parser functional	**4 / 4**	**PASSED**
TOTAL	**Comprehensive Full-Stack Test Suite**	**30 / 30**	**100% PASS**

8. Next Steps & Novel Research Contributions (Phase 8+)

With the Ploutos baseline successfully reproduced and fine-tuned, subsequent phases focus on novel extensions:

- **Phase 8 (Adaptive Expert Router):** Context-gated dynamic weighting network based on market volatility and regime.
- **Phase 9 (Visual Financial Expert):** Multimodal price-chart representations and visual feature ablation.
- **Phase 10 (Market-Regime Robustness):** Stress-testing across bull, bear, and high-volatility regimes.